Assignment : 4

1. What's the difference between .apk and .aab file? Create .apk and .abb file of your application.
2. What is Hive in Flutter why use?
3. What Fingerprint Does? How does fingerprint authentication work for individual users in a Flutter app? What's unique per user? How Can We Use It in an App With Many Users?

Question 1 : First: The Theory Part

Difference between .apk and .aab:

Feature	**.APK**	**.AAB**
Full Name	Android Package Kit	Android App Bundle
Purpose	Direct install on any device	Upload to Google Play Store
Size	Larger (contains everything)	Smaller (Play Store optimizes it)
Who uses it	Developers for testing/sharing	Google Play Store publishing
Install directly?	Yes	No (only via Play Store)

Simple explanation: APK is like a complete ready-made package you can install directly on any phone. AAB is a smarter format you give to Google Play — Google then creates the right APK for each specific phone automatically.

Now the Practical : Creating Both Files

We will use our **same payment_app** project from Assignment 3.

Step 1 : Create .APK file:

In VS Code terminal, inside payment_app folder:

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

PS D:\MAD_Lab\payment_app> flutter build apk --release
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS D:\MAD_Lab\payment_app> flutter build apk --release
Running Gradle task 'assembleRelease'...                        \
```

APK Created Successfully!

```
Running Gradle task 'assembleRelease'...                    1239.5s
√ Built build\app\outputs\flutter-apk\app-release.apk (52.8MB)
PS D:\MAD_Lab\payment_app>
```

Step 2 : Now Create .AAB file:

```
PS D:\MAD_Lab\payment_app> flutter build appbundle --release
Running Gradle task 'bundleRelease'...                          |
```

AAB Created Successfully!

```
PS D:\MAD_Lab\payment_app> flutter build appbundle --release
Running Gradle task 'bundleRelease'...                          49.8s
√ Built build\app\outputs\bundle\release\app-release.aab (51.7MB)
PS D:\MAD_Lab\payment_app>
```

Now Showing Both Files in Explorer:

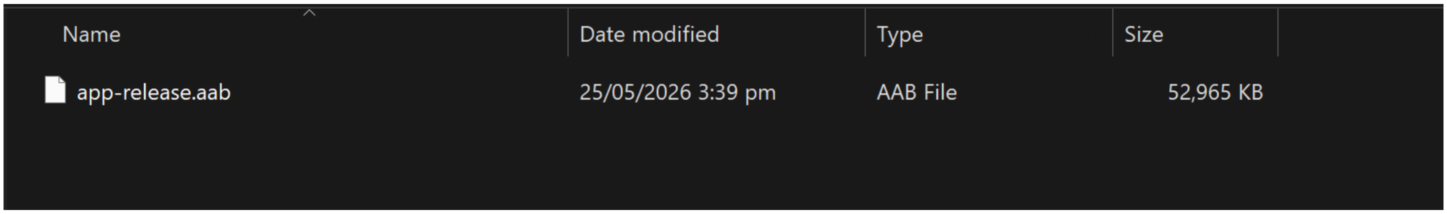
1. **APK file location:**

D:\MAD_Lab\payment_app\build\app\outputs\flutter-apk\

Name	Date modified	Type	Size
app-debug.apk	24/05/2026 12:06 am	APK File	98,882 KB
app-debug.apk.sha1	24/05/2026 12:06 am	SHA1 File	1 KB
app-release.apk	25/05/2026 3:31 pm	APK File	54,091 KB
app-release.apk.sha1	25/05/2026 3:31 pm	SHA1 File	1 KB

2. AAB file location:

D:\MAD_Lab\payment_app\build\app\outputs\bundle\release\



Question 2 : What is Hive in Flutter?

Theory Answer:

What is Hive?

Hive is a **lightweight and fast local database** for Flutter. It stores data directly on the device (offline) without needing internet or a server.

Why Use Hive?

Reason	Explanation
Very Fast	Faster than SQLite and other databases
Lightweight	Very small size, doesn't make app heavy
Works Offline	No internet needed, data saved on device
Easy to Use	Simple code, easy to learn
Secure	Supports encryption (AES-256)
Flutter Native	Made specifically for Flutter/Dart

Real Life Example:

Think of Hive like a **notebook on your phone** — you write something in it, close the app, open again — your data is still there!

Code Example

```
// Step 1: Add hive_flutter in pubspec.yaml
// hive_flutter: ^1.1.0

// Step 2: Initialize Hive
await Hive.initFlutter();
var box = await Hive.openBox('myBox');

// Step 3: Save data
box.put('name', 'Rifaq');
box.put('age', 22);

// Step 4: Read data
String name = box.get('name'); // Output: Rifaq

// Step 5: Delete data
box.delete('name');
```

Question 3 : Fingerprint Authentication

Theory Answer:

1. What Does Fingerprint Do?

Fingerprint authentication **verifies a user's identity** using their unique fingerprint pattern instead of a password. It is fast, secure and easy to use.

2. How Does Fingerprint Work?

The process is simple:

Step 1 — Registration: User puts finger on sensor → Phone scans and saves fingerprint pattern

Step 2 — Authentication: User puts finger again → Phone compares with saved pattern → If matched = Access Granted

What is Unique Per User?

What	Why Unique
Fingerprint Pattern	Every human has different ridges and lines
Stored Template	Saved as encrypted math template, not actual image
Device Secure Storage	Stored inside phone's secure chip (TEE)

How Can We Use It in App With Many Users?

Step	Explanation
Each user logs in with username/password first	App identifies WHO the user is
Then fingerprint is enabled for THAT user	Fingerprint just replaces password next time
Fingerprint never leaves the phone	It's always verified by the phone hardware
App only gets YES or NO from phone	Phone does all fingerprint matching

Simple Real Life Example:

Think of it like this — your phone knows YOUR finger. When you open a banking app, the app asks your phone "is this the right person?" — your phone checks the finger and just says YES or NO. The app never sees your actual fingerprint!

Now Practical : Fingerprint in Flutter:

Step 1 : Create New Project:

In VS Code Terminal (Run One After One)

- cd D:/MAD_Lab
- flutter create fingerprint_app
- cd fingerprint_app
- code .

Step 2 : Add Fingerprint Package

In VS Code terminal we will run:

- flutter pub add local_auth

```
PS D:\MAD_Lab\fingerprint_app> flutter pub add local_auth
```

Then:

- flutter pub get

```
PS D:\MAD_Lab\fingerprint_app> flutter pub get
```

Packages install successfully

```
PS D:\MAD_Lab\fingerprint_app> flutter pub add local_auth
Resolving dependencies...
Downloading packages... (3.1s)
+ flutter_plugin_android_lifecycle 2.0.34
+ intl 0.20.2
+ local_auth 3.0.1
+ local_auth_android 2.0.8
+ local_auth_darwin 2.0.3
+ local_auth_platform_interface 1.1.0
+ local_auth_windows 2.0.1
  matcher 0.12.19 (0.12.20 available)
  meta 1.17.0 (1.18.2 available)
+ plugin_platform_interface 2.1.8
  test_api 0.7.10 (0.7.12 available)
  vector_math 2.2.0 (2.3.0 available)
Changed 8 dependencies!
4 packages have newer versions incompatible with dependency constraints.
Try `flutter pub outdated` for more information.
PS D:\MAD_Lab\fingerprint_app> flutter pub get
Resolving dependencies...
Downloading packages...
  matcher 0.12.19 (0.12.20 available)
  meta 1.17.0 (1.18.2 available)
  test_api 0.7.10 (0.7.12 available)
  vector_math 2.2.0 (2.3.0 available)
Got dependencies!
4 packages have newer versions incompatible with dependency constraints.
Try `flutter pub outdated` for more information.
PS D:\MAD_Lab\fingerprint_app>
```

Step 3 : Configure Android Permission

Only 2 Things to Do:

1 — In AndroidManifest.xml:

android → app → src → main → AndroidManifest.xml

Find this line:

<manifest xmlns:android="http://schemas.android.com/apk/res/android">

Add this line right below it:

<uses-permission android:name="android.permission.USE_BIOMETRIC"/>

```
<manifest xmlns:android="http://schemas.android.com/apk/res/android">
<uses-permission android:name="android.permission.USE_BIOMETRIC"/>
```

2 — In MainActivity.kt:

android → app → src → main → kotlin → com → example → fingerprint_app → MainActivity.kt

Replace entire content with:

package com.example.fingerprint_app

import io.flutter.embedding.android.FlutterFragmentActivity

class MainActivity: FlutterFragmentActivity()

android > app > src > main > kotlin > com > example > fingerprint_app > MainActivity.kt

```
package com.example.fingerprint_app

import io.flutter.embedding.android.FlutterFragmentActivity

class MainActivity: FlutterFragmentActivity()
```

Step 4 : Write the Fingerprint Code

Updated main.dart

```dart
import 'package:flutter/material.dart';
import 'package:local_auth/local_auth.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Fingerprint App',
      debugShowCheckedModeBanner: false,
      theme: ThemeData(
        colorScheme: ColorScheme.fromSeed(seedColor: Colors.indigo),
        useMaterial3: true,
      ), // ThemeData
      home: const FingerprintScreen(),
    ); // MaterialApp
  }
}

class FingerprintScreen extends StatefulWidget {
  const FingerprintScreen({super.key});

  @override
  State<FingerprintScreen> createState() => _FingerprintScreenState();
}

class _FingerprintScreenState extends State<FingerprintScreen> {
  final LocalAuthentication auth = LocalAuthentication();
  String statusMessage = 'Press button to authenticate';
  bool isAuthenticated = false;

  Future<void> authenticate() async {
    try {
      bool canCheck = await auth.canCheckBiometrics;
      if (!canCheck) {
        setState(() {
          statusMessage = '❌ Fingerprint not available on this device';
        });
        return;
      }

      bool result = await auth.authenticate(
        localizedReason: 'Please scan your fingerprint to continue',
      );

      setState(() {
        isAuthenticated = result;
        statusMessage = result
            ? '✅ Authentication Successful! Welcome Rifaq!'
            : '❌ Authentication Failed! Try Again.';
      });
    } catch (e) {
      setState(() {
        statusMessage = '❌ Error: $e';
      });
    }
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Fingerprint Authentication'),
        backgroundColor: Colors.indigo,
        foregroundColor: Colors.white,
      ), // AppBar
      body: Padding(
        padding: const EdgeInsets.all(24.0),
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: [
            Icon(
              isAuthenticated ? Icons.lock_open : Icons.fingerprint,
              size: 120,
              color: isAuthenticated ? Colors.green : Colors.indigo,
            ), // Icon
            const SizedBox(height: 30),
            Text(
```

```dart
                statusMessage,
                textAlign: TextAlign.center,
                style: TextStyle(
                  fontSize: 18,
                  fontWeight: FontWeight.bold,
                  color: statusMessage.contains('✅')
                      ? Colors.green
                      : statusMessage.contains('❌')
                          ? Colors.red
                          : Colors.black,
                ), // TextStyle
              ), // Text
              const SizedBox(height: 40),
              SizedBox(
                width: double.infinity,
                height: 55,
                child: ElevatedButton.icon(
                  onPressed: authenticate,
                  icon: const Icon(Icons.fingerprint),
                  label: const Text(
                    'Authenticate with Fingerprint',
                    style: TextStyle(fontSize: 16),
                  ), // Text
                  style: ElevatedButton.styleFrom(
                    backgroundColor: Colors.indigo,
                    foregroundColor: Colors.white,
                    shape: RoundedRectangleBorder(
                      borderRadius: BorderRadius.circular(12),
                    ), // RoundedRectangleBorder
                  ),
                ), // ElevatedButton.icon
              ), // SizedBox
              const SizedBox(height: 20),
              if (isAuthenticated)
                Container(
                  padding: const EdgeInsets.all(16),
                  decoration: BoxDecoration(
                    color: Colors.green.shade50,
                    borderRadius: BorderRadius.circular(12),
                    border: Border.all(color: Colors.green),
                  ), // BoxDecoration
                  child: const Row(
```

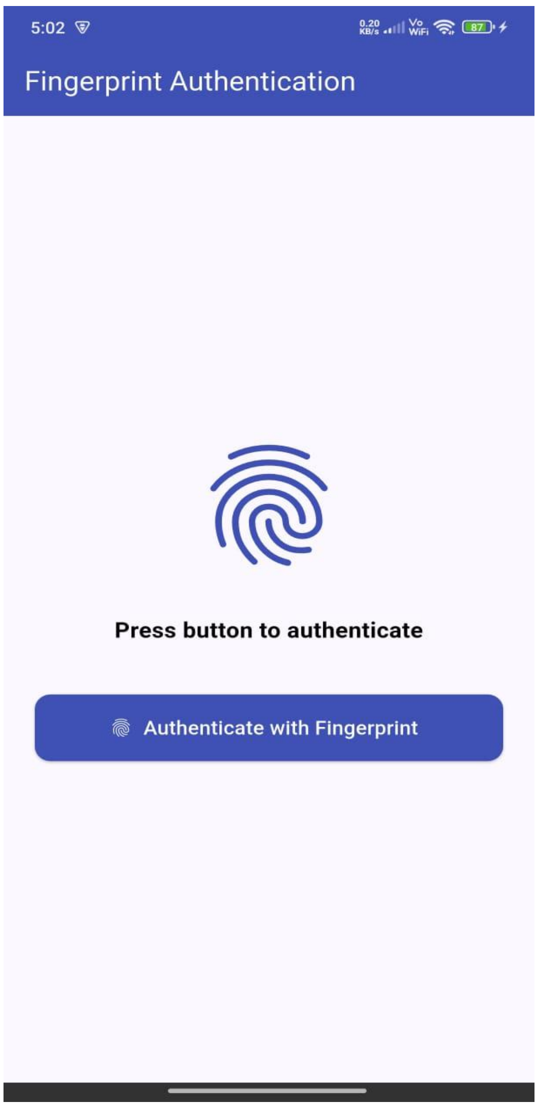
```dart
                    children: [
                      Icon(Icons.check_circle, color: Colors.green),
                      SizedBox(width: 10),
                      Text(
                        'Access Granted!',
                        style: TextStyle(
                          color: Colors.green,
                          fontWeight: FontWeight.bold,
                          fontSize: 16,
                        ), // TextStyle
                      ), // Text
                    ],
                  ), // Row
                ), // Container
            ],
          ), // Column
        ), // Padding
      ); // Scaffold
    }
  }
```

Final Step : Running

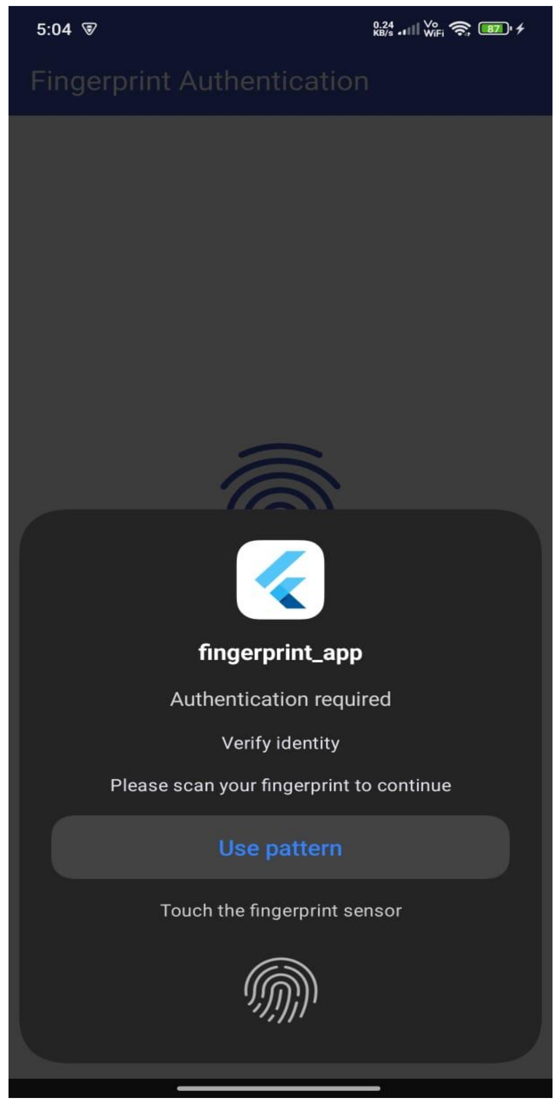
flutter run -d 23088PND5R

```
Running Gradle task 'assembleDebug'...
```

Running Successfully



1. The fingerprint scan popup appearing



2. Authentication Successful! Welcome Rifaq!